

CPSC 4300/6300: Assignment 5

Instructions

- Please submit your solution by 11:59PM December 5. **There will be no extensions for this assignment.**
- Submissions should be made through **Canvas**.
- Please complete homework individually.
- Please submit a **single PDF/Word file** including your solutions and the code of your solutions. Feel free to use any third-party libraries.

This assignment has two major parts:

Part 1. Image Classification using ANN and CNN (STL-10 dataset)

Part 2. NLP + Large Language Model Concepts (conceptual questions)

You must follow all architecture specifications exactly.

Part 1: Image Classification (80 points)

The dataset used for this part is STL-10, for more details, check [here](#):

1. (20 points) Before training your models, apply the following preprocessing steps to every image:

- 1) Resize each image to 64×64 pixels
- 2) Convert the image to a tensor
- 3) Normalize the image using: mean = (0.4467, 0.4398, 0.4066); standard deviation = (0.2241, 0.2215, 0.2239)

2.(30 points) ANN Architecture (Must Implement This Exactly)

After preprocessing the image, your ANN must take the flattened image as input:

- An RGB image of size $3 \times 64 \times 64$
- Flattened into 12,288 features

A fully connected neural network with the following layers is required:

1) Layer 1:

- Input dimension: 12,288
- Output dimension: 1,024
- Activation: ReLU

2) Layer 2:

- Input dimension: 1,024
- Output dimension: 512
- Activation: ReLU

3) Layer 3:

- Input dimension: 512
- Output dimension: 10 (one score for each STL-10 class)

CPSC 4300/6300: Assignment 5

Training Requirements

- Train for 5 epochs
- Loss function: CrossEntropyLoss
- Optimizer: Adam, learning rate = 0.001

3. (30 points) Build and Train CNN

Input: $3 \times 64 \times 64$ color image

1) Layer 1 Convolution

- Input channels: 3
- Output channels: 32
- Kernel: 5×5
- Padding: 2
- Activation: ReLU

2) Max Pooling

- Kernel: 2
- Stride: 2

3) Layer 2 Convolution

- Input channels: 32
- Output channels: 64
- Kernel: 3×3
- Padding: 1
- Activation: ReLU

4) Layer 3 Convolution

- Input channels: 64
- Output channels: 128
- Kernel: 3×3
- Padding: 1
- Activation: ReLU

5) Max Pooling

- Kernel: 2
- Stride: 2

6) Flatten to 32,768 features

7) Fully connected layer

- Input: 32,768
- Output: 512
- Activation: ReLU

8) Fully connected output layer

- Input: 512
- Output: 10

CPSC 4300/6300: Assignment 5

Training Requirements:

- Train for 15 epochs
- Loss: CrossEntropyLoss
- Optimizer: Adam, learning rate = 0.0001

Deliverable:

- The training loss for each epoch
- Compare the test accuracy of ANN vs CNN
- Your code showing the steps
- A short explanation
 - Why did the ANN struggle?
 - Why did the CNN perform better?
 - Which architectural components made the biggest difference?

Part 2: NLP + LLM (20 points)

4. Explain the preprocessing steps used in NLP.
5. What is the Distributional Hypothesis?
6. From the reading material, the authors mention that LLMs can display emergent behaviors. Explain what emergent behaviors are and give one example.
7. From the reading material, using the sentence “The tiger jumped out of a tree... because it was thirsty”, explain how self-attention links “it” to “the tiger.”